

Anjila Belbase

Tampa, FL, USA | 813-573-3270 | belbaseanjila@gmail.com | www.linkedin.com/in/anjilab

EDUCATION

University of South Florida

Bachelor of Science in Electrical Engineering

Tampa, FL

December 2025

EXPERIENCE

Controls Engineering Specialist

August 2024 – December 2025

Cirkul Inc.

Tampa, FL

- Developed detailed Bills of Materials (BOMs), wiring schematics, and SOPs using AutoCAD to standardize build processes, driving a 25% reduction in rework and documentation errors.
- Engineered ladder logic and structured text for Allen-Bradley PLCs and built HMI screens to stabilize automated equipment, improving production uptime by 10%.
- Resolved chronic equipment failures by applying root-cause troubleshooting with cross-functional teams and implementing targeted PLC logic corrections, boosting reliability by 15%.
- Utilized Studio 5000, RSLogix 500, Productivity Suite, FactoryTalk View, and C-More HMI while integrating basic Cognex vision tools to enhance inspection and quality workflows.
- Led multiple controls initiatives, including:
 - Industrial Dryer and Washer Redesign: Commissioned redesigned wash/dry systems by validating I/O, integrating safety interlocks, and tuning control logic, reducing cycle variability by 10%.
 - Real-Time Status and Alarm Platform: Built a PLC-based status and alarm interface that strengthened operator visibility and cut troubleshooting delays by 15%.
 - Automated Maintenance and Diagnostics Interface: Streamlined maintenance by automating sensor calibration routines and creating a real-time diagnostics HMI dashboard, accelerating issue resolution by 20%.

Electrical Engineering Intern

May 2024 – August 2024

Cirkul Inc.

Tampa, FL

- Absorbed manufacturing and automation workflows by shadowing controls engineers and industrial electricians, reducing ramp-up time by 30%.
- Supported large-scale controls upgrades by updating AutoCAD schematics, assisting with field wiring, modifying PLC logic, and preparing SOPs, improving project throughput by 15%.
- Explored new automation solutions through hands-on testing, research, and prototype trials, strengthening feasibility assessments for future process improvements.

TECHNICAL SKILLS

Automation & Controls: Allen-Bradley Controls, Studio 5000, RSLogix 500, FactoryTalk View, Productivity Suite, C-More

Programming & Simulation: Python, C, C++, MATLAB, Ladder Logic / PLC Programming

Design & Documentation: AutoCAD, Fusion 360 (CAD Software), 3D printing, BOMs, SOPs,

PROJECTS

Smart IoT Farming System for Microgreens

January 2025 – December 2025

Tools: ESP32, C++, MQTT, Wi-Fi, NuvIoT Cloud, Fusion 360 (3D Printing)

- Collaborated with a 3-member team to build an ESP32-based IoT platform that streamed live temperature, humidity, CO2, soil moisture, gas pressure and light data to the NuvIoT cloud every 30s via MQTT, achieving >98% uptime during prototype testing.
- Led software and MQTT integration with structured multi sensor payloads and optimized send timing, achieving >98% transmission reliability and sub second cloud update latency.
- Designed and 3-D printed a custom enclosure inspired by a ladybug form factor, improving sensor protection and durability in humid environments.

JamGesture: IMU-Based Gesture-Controlled MP3 System

March 2025 – May 2025

Tools: ESP32, IMU, C, Kalman Filter, DFPlayer Mini, Fusion 360 (3D Printing)

- Programmed the ESP32 to map unique hand gestures to playback and volume actions, achieving >95% recognition accuracy during testing.
- Improved IMU stability by 60% using Kalman filtering, reducing false triggers and enabling smooth real-time response.
- Created a 3D-printed wrist-mounted housing, reducing electronic noise from motion by 35% and improving wearability for users.